

**MAE 4021 Senior Design Project Wind Turbine Blades:
Fall 2025 Prof. R.J. Barthelmie & Dr R. Bhaskaran**

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Abstract

Small-scale wind turbines operate in a challenging aerodynamic regime where low Reynolds numbers, rapid stall onset, and strong geometric sensitivities make blade design particularly complex. Understanding and improving the performance of these turbines requires an integrated approach that connects fundamental aerodynamic modeling, high-fidelity numerical simulation, and controlled experimental testing. This project investigates the aerodynamic behavior of a three-bladed horizontal-axis wind turbine using the NACA 2414 airfoil, with the broader goal of developing a structured methodology for analyzing and improving blade performance within the constraints of the MAE 4021 wind tunnel. The work combines Blade Element Momentum (BEM) theory, computational fluid dynamics (CFD) modeling in ANSYS Fluent, and physical wind-tunnel measurements to examine how geometry, specifically chord distribution, twist distribution, and pitch, shapes turbine efficiency across a range of tip-speed ratios. Through this multi-modal framework, the study aims to build a deeper understanding of small-turbine aerodynamics, evaluate the capabilities and limitations of computational and experimental tools, and establish a foundation for future optimization, structural analysis, and validation efforts.

Executive Summary

Desired Functions:

The goal of this project is to design and optimize a small-scale wind turbine capable of extracting the maximum possible aerodynamic power from the wind under low-Reynolds number conditions. The blade must efficiently convert wind energy into mechanical shaft power by achieving a high power coefficient C_p while maintaining stable aerodynamic behavior over a practical operating range of tip speed ratios. The design must also remain compatible with the NACA 2414 airfoil and the physical constraints of the wind-tunnel experiment.

Constraints Related to Main Functions:

Several constraints govern the design. The rotor radius is fixed at 0.305 m to match wind-tunnel hardware limitations, and a small hub cutout is required for structural clearance. The geometry must be manufacturable, so chord and twist values cannot be excessively large or small, and a relatively simple, linear taper is preferred for fabrication. Aerodynamically, the design must keep the angle of attack within reliable limits of the available NACA 2414 airfoil data, and it must operate effectively at wind-tunnel speeds between 5 and 7 m/s without stalling. Additionally, the blade must use the NACA 2414 airfoil profile consistently along its span.

Performance Objectives:

The primary objective is to optimize the blade design to achieve a power coefficient higher than the standard blade's wind-tunnel performance as given by test data, which peaks

around $C_p \approx 0.43$ at $\lambda \approx 6.1$. The optimized blade should also demonstrate a clear and stable optimal tip-speed ratio between 4 and 8, provide well-behaved chord and twist distributions, and avoid aerodynamic stall. Quantitatively, the design seeks to maximize C_p , identify the corresponding λ_{opt} , and output the full radial distributions $c(r)$ and $\theta_p(r)$ that define the improved geometry.

Alternative Design Concepts:

Three classes of designs were evaluated. First, the baseline “standard blade” with constant chord and constant pitch was analyzed to replicate wind-tunnel ANSYS Fluent Results. Second, an optimum-rotor blade derived from Manwell’s rotor theory was explored, using the angle of attack that maximizes C_L/C_D to define an idealized chord and twist distribution. Third, a family of modified geometries was examined by scaling chord length, adjusting root twist, and applying linear twist gradients along the radius. These variations formed the search space for identifying an improved blade geometry.

Analyses Applied in Selecting Design Concepts:

Selection among design concepts relied on a combination of analytical and numerical tools. The Blade Element Momentum (BEM) Method, implemented in MATLAB with Prandtl tip-loss and Glauert high-thrust corrections, evaluated $C_p(\lambda)$ for thousands of candidate geometries. ANSYS Fluent simulations provided detailed flow-field and torque predictions for the standard blade, enabling comparison against the BEM model. Experimental wind-tunnel data for the standard blade served as ground truth for validation. Together, these analyses facilitated systematic comparison of aerodynamic performance, stability, and manufacturability.

Relevant Industry/Society Standards:

The design draws on relevant engineering standards and accepted industry practices. The IEC 61400-2 standard for small wind turbines guided considerations of safety, operating conditions, and performance evaluation. Airfoil data from AirfoilTools, widely used in aerodynamic analysis, provided authoritative NACA 2414 lift and drag characteristics. The methodology in Manwell, McGowan, and Rogers’ *Wind Energy Explained* served as the standard framework for rotor optimization and BEM implementation.

Concepts Derived from Coursework:

This design integrates several concepts from prior coursework. From MAE 4021 (Wind Power Systems), the project uses BEM theory, optimum-rotor design, and power-coefficient analysis. Fluid mechanics from MAE 3240 informs understanding of lift, drag, stall, and Reynolds-number effects. Dynamics concepts from MAE 2030 support torque and rotational power analysis. Fluid analysis principles from MAE 4272 supported understanding of mesh generation and simulation in ANSYS Fluent. Computational problem-solving and programming skills from CS 1112 (MATLAB) and MATLAB-based courses support data interpolation, iterative solvers, and optimization loops. Together, these courses form the technical foundation of the design.

Design Evaluation & Comparison to Existing Solutions:

The optimized blade design meets its primary objective of improving aerodynamic efficiency while satisfying all geometric and manufacturing constraints. By adjusting the chord and twist distributions using the BEM framework, the redesigned blade achieves a higher predicted power coefficient than the standard NACA 2414 blade used in the wind-tunnel turbine, with a peak C_p of about 0.44 compared to the standard blade's ~ 0.42 . The resulting geometry remains smooth, manufacturable, and structurally reasonable, avoiding extreme taper or excessive twist that could create fabrication or loading issues. Importantly, the optimized blade operates most efficiently at a slightly lower tip-speed ratio, which may benefit longevity by reducing tip erosion, noise, and fatigue loads, though it could also influence generator matching or startup characteristics. Compared with existing small-turbine designs, the optimized blade follows well-established trends - higher loading near the root and reduced twist toward the tip - indicating that the method produces physically realistic results. While full validation would require finer CFD, structural analysis, and prototype testing, the optimized blade demonstrates a clear performance gain within the limitations of the BEM approach and provides a strong foundation for future refinement.

Design Impacts:

The optimized blade design has the potential to generate meaningful positive impacts on public health, safety, and human welfare by enabling more efficient small-scale wind energy systems. Improved turbine performance at lower tip-speed ratios reduces noise emissions and blade-tip erosion, creating safer and more community-friendly operation, which is an important factor for distributed wind installations near residential or agricultural areas. Environmentally, even modest gains in C_p translate into higher energy output per unit swept area, which can reduce dependence on fossil fuels and lower lifecycle carbon emissions. Culturally and socially, accessible and efficient small turbines support energy resilience for rural communities, off-grid users, and developing regions where reliable electrical infrastructure is limited. Economically, a more productive rotor increases the return on investment for small wind systems, lowering the cost per kilowatt-hour and making renewable energy more competitive. While the design remains conceptual until validated through structural analysis and field testing, its implementation aligns strongly with global sustainability goals by promoting cleaner energy generation, reducing environmental impact, and improving energy accessibility across diverse communities.

Design & Results Format:

The final design took the form of an analytical and computational model rather than a manufactured prototype. Specifically, the work produced a fully parameterized Blade Element Momentum (BEM) implementation in MATLAB capable of generating optimized chord and twist distributions for a fixed-radius NACA 2414 rotor. The output includes detailed numerical geometry tables, performance predictions across a wide range of tip-speed ratios, and direct comparisons against the experimentally measured “standard blade.” In parallel, a complementary

CFD modeling framework was developed in ANSYS Fluent, which enabled independent aerodynamic validation of several operating points through torque extraction and pressure-field visualization. While the project did not produce CAD drawings or a physical blade, the combination of optimized BEM geometry, validated simulation workflows, and standardized analysis scripts constitutes a complete digital design package that could directly support future fabrication, prototyping, or structural evaluation.

Project Roles:

This project was completed individually. Responsibilities included implementing the BEM solver, performing geometry parameter sweeps, analyzing and validating NACA 2414 airfoil data, running CFD simulations of the standard blade in ANSYS Fluent, processing wind-tunnel data, and selecting the optimized geometry. All MATLAB programming, CFD setup, data visualization, and report preparation were carried out independently.

Flow Around Standard Blade Design Using Ansys Fluent

Introduction

The turbine analyzed in this project is a three-bladed, horizontal-axis wind turbine with a radius of 0.305 m and a constant NACA 2414 airfoil profile along the entire span. The blade includes a variable chord distribution that tapers toward the tip and operates at pitch angles ranging from approximately $+18.9^\circ$ to -2° . Because structural dimensions, laminate layups, or material properties were not provided, a structural finite element analysis was beyond the scope of this project. We assume the blade remains rigid under the aerodynamic loads considered during CFD testing, allowing us to focus entirely on fluid–structure aerodynamic interactions.

The computational fluid dynamics (CFD) model was developed in ANSYS Fluent. This involved importing blade geometry as a surface body and generating a surrounding 3D flow volume. The volume was made sufficiently large that the velocity at the far boundaries remained nearly uniform, which was consistent with the assumption that sufficiently far from the rotor, the flow returns to undisturbed freestream conditions. After defining the fluid region, the blade was rotated in SpaceClaim to achieve the desired pitch angle for each case (e.g., 4° or 0°), and a mesh was generated for each configuration. The CFD simulation then applied a rotating reference frame corresponding to each tested tip-speed ratio (TSR), and the model was run to convergence to obtain torque and, ultimately, the power coefficient C_p .

Governing Equations

The flow around the rotating NACA 2414 blade was modeled in a rotating reference frame (RRF), so that the blade appears stationary and the fluid “moves past it” in the rotating frame. In this formulation the absolute velocity $\mathbf{v}$ is decomposed as

$$\mathbf{v} = \mathbf{v}_r + \boldsymbol{\omega} \times \mathbf{r}$$

where $\mathbf{v}_r$ is the relative velocity in the rotating frame, $\boldsymbol{\omega}$ is the rotor angular velocity vector, and $\mathbf{r}$ is the position vector from the rotation axis.

Under this rotating-frame assumption, ANSYS Fluent solves the steady, incompressible, Reynolds-Averaged Navier-Stokes (RANS) equations for $\mathbf{v}_r$ and p :

- Continuity (mass conservation)

For constant-density air:

$$\nabla \cdot \mathbf{v}_r = 0$$

- Momentum (RANS in rotating frame)

$$\rho(\mathbf{v}_r \cdot \nabla) \mathbf{v}_r + \rho[2\boldsymbol{\omega} \times \mathbf{v}_r + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})] = -\nabla p + \mu \nabla^2 \mathbf{v}_r + \nabla \cdot \boldsymbol{\tau}_{turb}$$

The extra terms in brackets represent the Coriolis acceleration $2\boldsymbol{\omega} \times \mathbf{v}_r$ and centripetal acceleration $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$ that arise when the equations are written in the rotating frame.

The last term, $\nabla \cdot \boldsymbol{\tau}_{turb}$ is the Reynolds stress contribution from turbulence modeling. In this project I used the SST $k - \omega$ model, so Fluent also solves transport equations for:

- k - turbulent kinetic energy
- ω - specific dissipation rate

These are coupled to the mean flow through the eddy viscosity $\mu_t = \rho a_1 k / \max(a_1 \omega, SF_2)$ (in Fluent's SST implementation), and they appear in the turbulent stress tensor closure $\boldsymbol{\tau}_{turb}$. At each cell center, Fluent therefore solves for six primary unknowns: the three components of $\mathbf{v}_r = (u, v, w)$, the static pressure p , and the turbulence variables k and ω . All other quantities (lift, drag, torque, C_p , etc.) are obtained in post-processing from these fields.

Because the equations are written in the rotating frame, the mesh does not need to move: the blade geometry is fixed in the computational domain, and rotation is captured entirely by the extra Coriolis and centripetal terms in the momentum equations.

Boundary Conditions

In the CFD model, the computational domain represented a 120° sector surrounding a single turbine blade. This reduction in domain size significantly decreased computational cost while still capturing the full three-bladed rotor behavior through azimuthal periodicity. The upstream face of the domain was assigned as a velocity inlet, where a uniform freestream wind

speed U_∞ was prescribed. All operating points in this study were evaluated at either 6 m/s or 5.5 m/s, depending on the specific pitch condition being validated against wind-tunnel data. The turbulence quantities at the inlet were specified using low-intensity values consistent with the SST $k - \omega$ model recommendations for external aerodynamic flows.

The downstream boundary was modeled as a pressure outlet with a static gauge pressure of zero Pascals, allowing disturbances in the wake to convect naturally out of the domain without artificial reflection. Because only a single blade was modeled, the two vertical faces bounding the sector were set as rotationally periodic, imposing a 120° phase shift to simulate the presence of the remaining two blades. This condition ensures that any flow structure exiting one periodic face re-enters through the other, maintaining the physical symmetry of a three-bladed rotor.

The blade surface itself was treated as a no-slip wall, enforcing zero relative velocity between the fluid and the blade in the rotating reference frame. Wall-adjacent turbulence quantities were handled using the SST-compatible near-wall formulation within Fluent. The outer cylindrical boundary enclosing the rotor was placed sufficiently far from the blade to avoid blockage effects and was treated as a slip or symmetry condition, permitting tangential flow while preventing normal penetration or shear. This boundary placement and treatment ensured that the flow near the blade was not artificially constrained by the outer walls, allowing a realistic development of the wake.

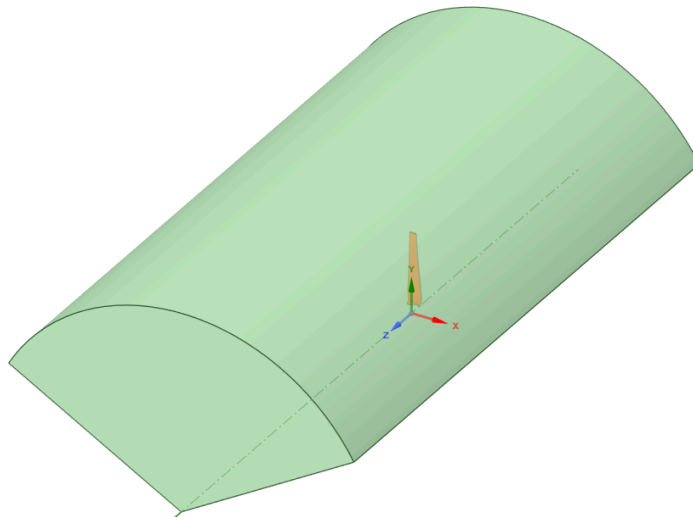


Figure 1: Blade Geometry and Solution Operating Volume

Numerical Solution Strategy

The CFD simulations were performed using the finite volume method implemented in ANSYS Fluent. The computational volume surrounding the blade was meshed with a combination of boundary-layer-refined cells near the blade surface and progressively coarser cells further away. This approach ensured adequate resolution of the leading-edge and trailing-edge flow gradients, tip vortices, and near-wall shear layers. Fluent's steady-state pressure-based RANS solver was employed, solving the incompressible Navier-Stokes equations and the two turbulence-transport equations of the SST $k - \omega$ model. Pressure-velocity coupling was achieved through a segregated simple-type algorithm, and spatial discretization was performed using upwind-biased schemes for stability.

At each blade pitch angle, the airfoil geometry was rotated in SpaceClaim to the appropriate orientation, requiring a new mesh for each configuration. For each tip-speed ratio examined, the corresponding angular velocity Ω was calculated using $\Omega = \frac{\lambda U_{\infty}}{R}$ and applied to the rotating reference frame. Unlike the strategy described in the course guidance, which recommends using the previous TSR's converged flow field as the initial condition for the next case, in this project I fully re-initialized the solution for each operating point. While this approach substantially increased the number of iterations required for convergence, it did not adversely affect the final results because steady RANS simulations of this type converge to the same solution regardless of initialization, provided that residuals and torque levels are driven to stable plateaus.

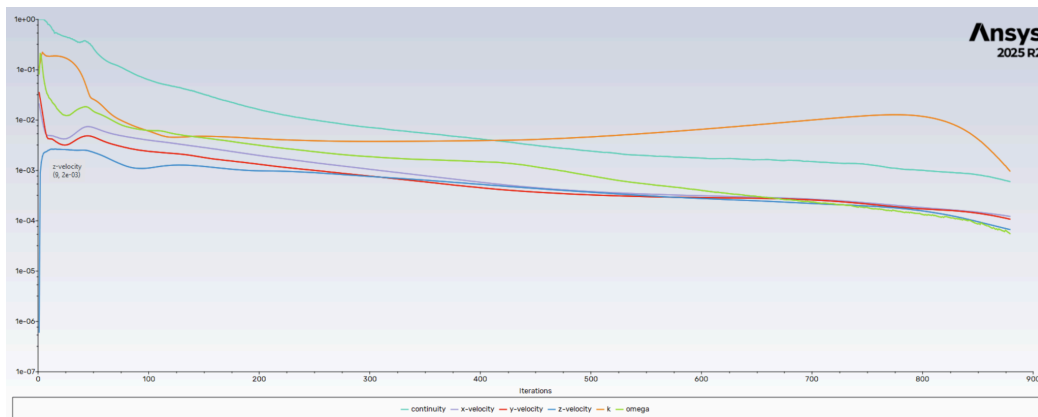


Figure 2: Scaled Residuals Convergence

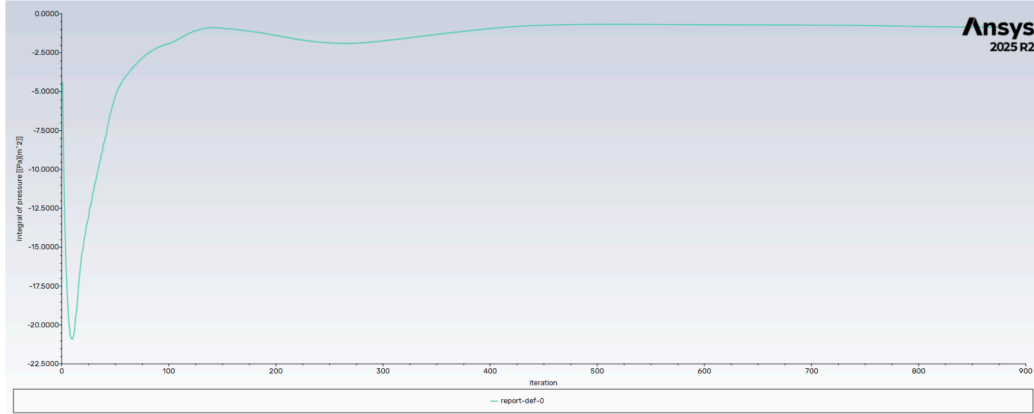


Figure 3: Pressure Integral Surface Report

Each simulation was run until the residuals of the continuity, momentum, and turbulence equations decreased significantly and the integrated torque on the blade reached a steady value. The torque predicted on the single blade was multiplied by three to obtain the total rotor torque. Rotor power was then computed from $P_{rotor} = B\tau_{blade}\Omega$, and the power coefficient was evaluated using

$$C_p = \frac{P_{rotor}}{P_{wind}} = \frac{B\tau_{blade}\Omega}{0.5\rho AU^3}$$

where $P_{wind} = 0.5\rho AU^3$ and $A = \pi R^2$ is the rotor swept area. These power-coefficient values formed the basis of the CFD-derived $C_p(\lambda)$ curves, which were subsequently compared to wind-tunnel results and to the BEM predictions developed in MATLAB.

Ansys Simulation of Standard Blade

Constant Operating Parameters:

- Radius = 0.305 m
- Standard NACA 2414 blade profile
- Relatively coarse mesh: 64,037 Cells

*Datasheet including all CFD results linked in Appendix B

$$U = 6 \text{ m/s}, \theta_p = 0^\circ$$

Simulation Results:

TSR λ_r	Angular Speed Ω [rad/s]	Torque [N·m]	Power of Rotor [W]	C_p
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2.5	49.180	0.0152832	2.2925	0.0593
3	59.016	0.0175855	3.1654	0.0819
4	78.688	0.0221729	5.225	0.1359
5	98.360	0.0274322	8.0651	0.2086
6	118.033	0.0263918	9.3427	0.2416
7	137.705	0.0235459	9.7480	0.2521
8	157.377	0.0146505	6.9004	0.1785
9	177.049	0.00189668	1.0071	0.0260

Power Coefficient vs. Lambda [λ], Pitch angle = 0 degrees, U=6 m/s

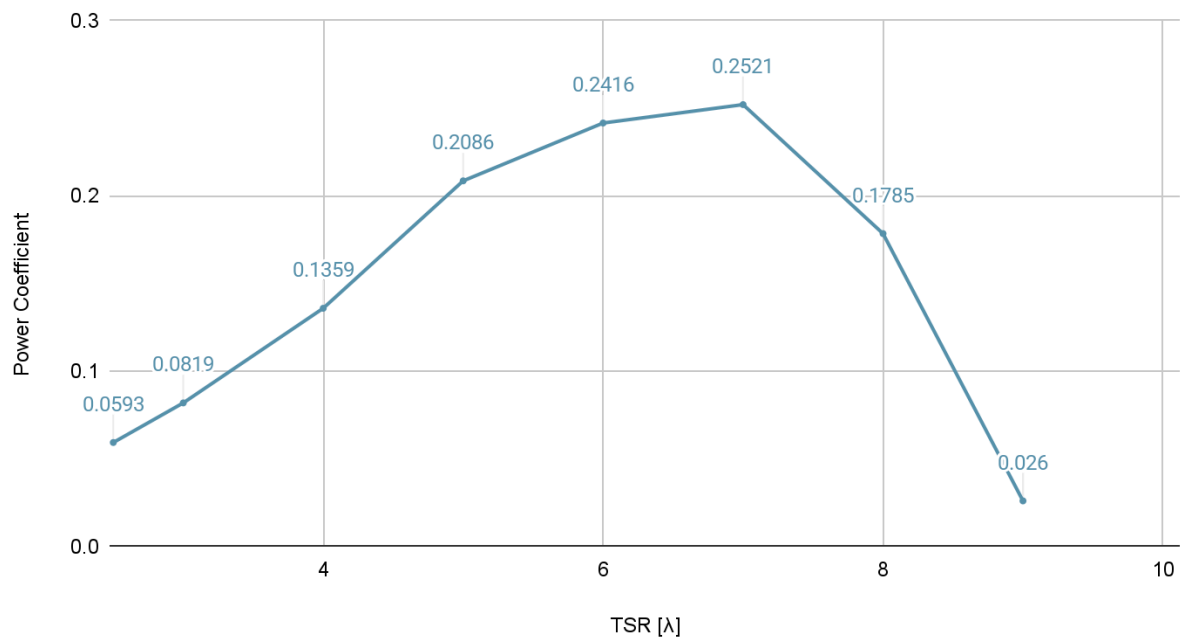
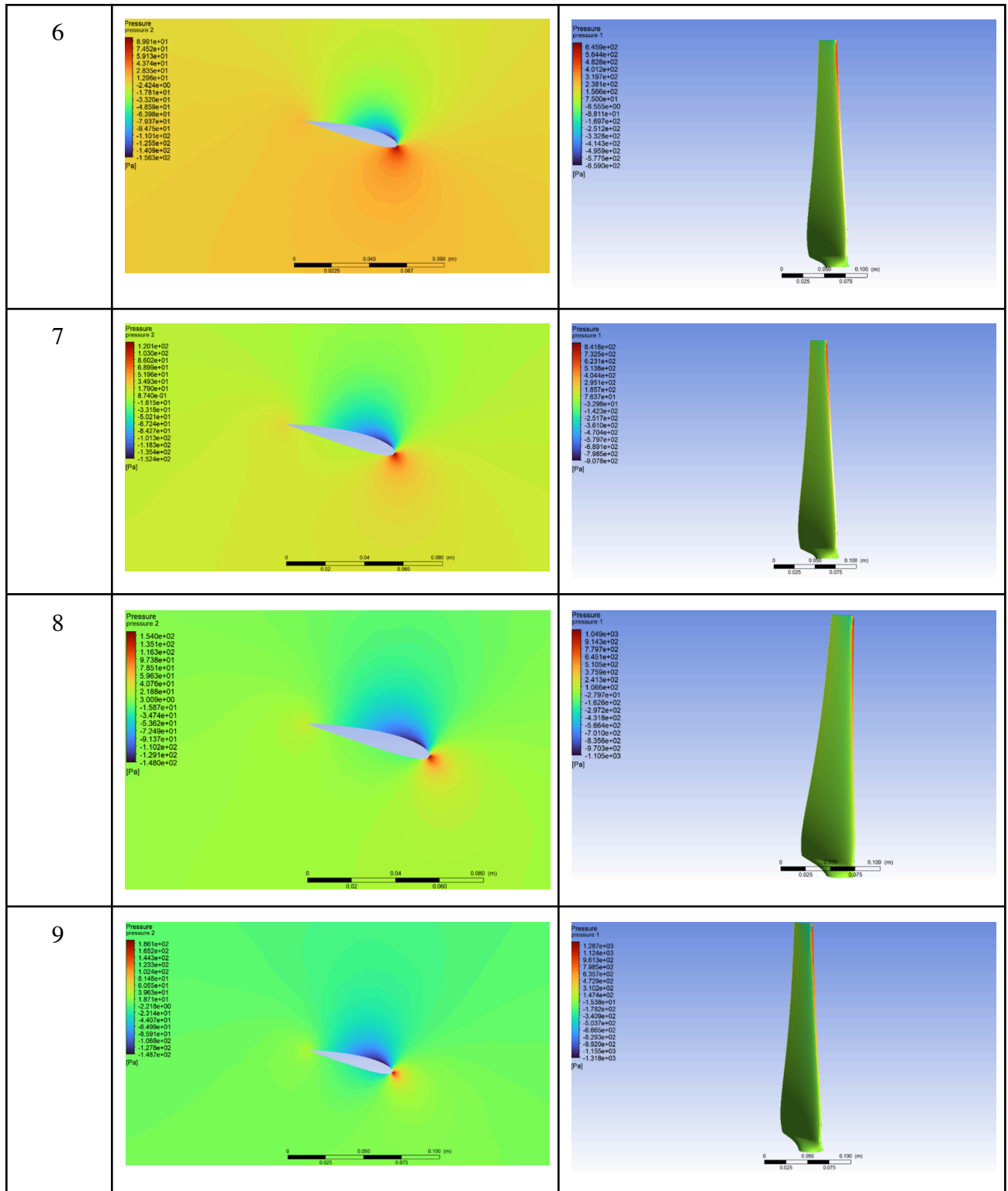


Figure 4: C_p vs λ from Ansys Fluent Data for U=6 m/s, $\theta_p = 0^\circ$

Pressure Distributions:

TSR λ_r	XZ Plane Pressure Distribution	Pressure Distribution Along Blade Surface
2.5		
3		
4		
5		



$$U = 6 \text{ m/s}, \theta_p = 4^\circ$$

Simulation Results:

TSR λ_r	Angular Speed Ω [rad/s]	Torque [N·m]	Power of Rotor [W]	C_p
2.5	49.180	0.0238175	3.5726	0.0924
3	59.016	0.0283717	5.1069	0.1321
4	78.688	0.0397437	9.4193	0.2436
5	98.360	0.0406077	11.9387	0.3088
6	118.033	0.0320382	11.3415	0.2933
7	137.705	0.0169852	7.0319	0.1819
8	157.377	0.000735713	0.3465	0.0090

Power Coefficient vs. Lambda [λ], Pitch angle = 4 degrees, U=6 m/s

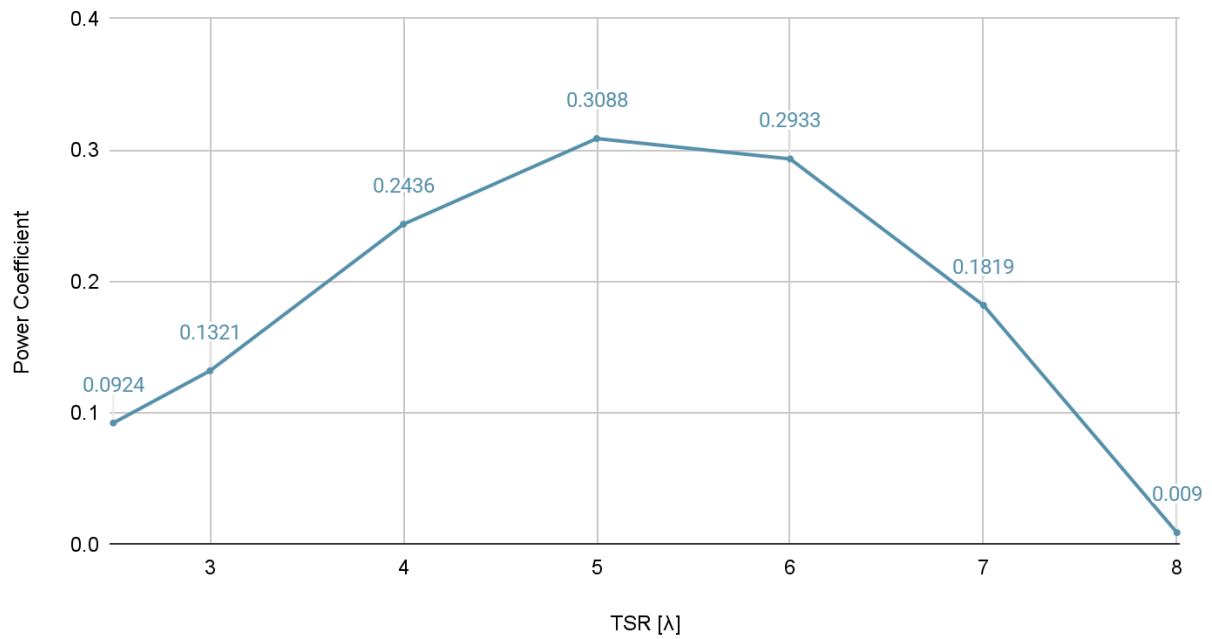
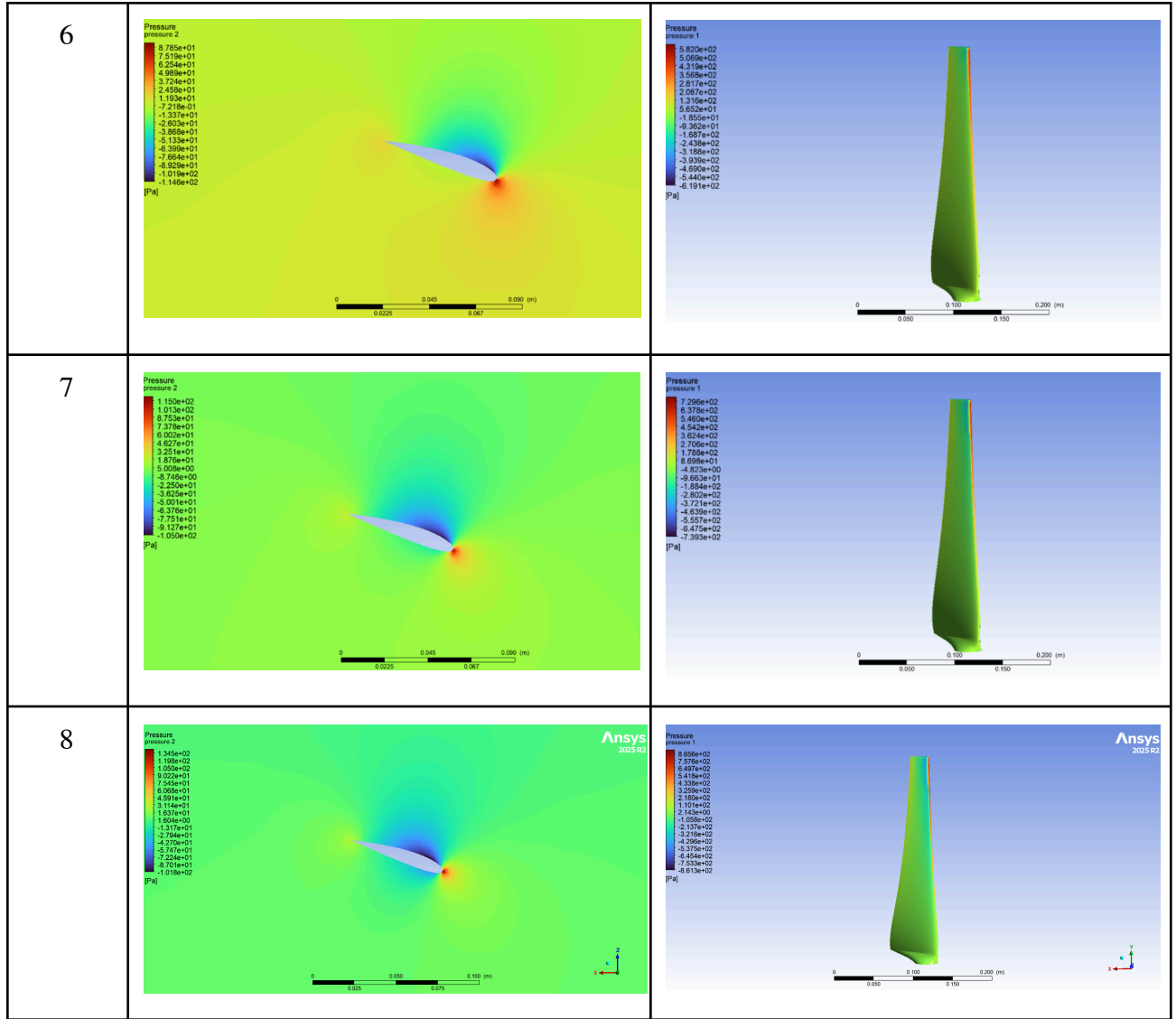


Figure 5: C_p vs λ from Ansys Fluent Data for U=6 m/s, $\theta_p = 4^\circ$

Pressure Distributions:

TSR λ_r	XZ Plane Pressure Distribution	Pressure Distribution Along Blade Surface
2.5		
3		
4		
5		



$$\underline{U = 5.5 \text{ m/s, } \theta_p = 0^\circ}$$

Simulation Results:

TSR λ_r	Angular Speed Ω [rad/s]	Torque [N·m]	Power of Rotor [W]	C_p
4.5	81.15	0.0218042	5.308071639	0.1782350549
5.6	101.16	0.0220317	6.686440362	0.2245180822
6.6	118.30	0.0214499	7.612253036	0.2556051293
7.4	133.26	0.0168689	6.743964989	0.2264496509

8.3	149.67	0.00864446	3.881504252	0.1303336071
9.6	173.111	-0.00624054	-3.240988643	-0.1088262985
10.1	182.31	-0.0127997	-7.000596575	-0.2350668566

Power Coefficient vs. Lambda [λ], Pitch angle = 0 degrees, U=5.5 m/s

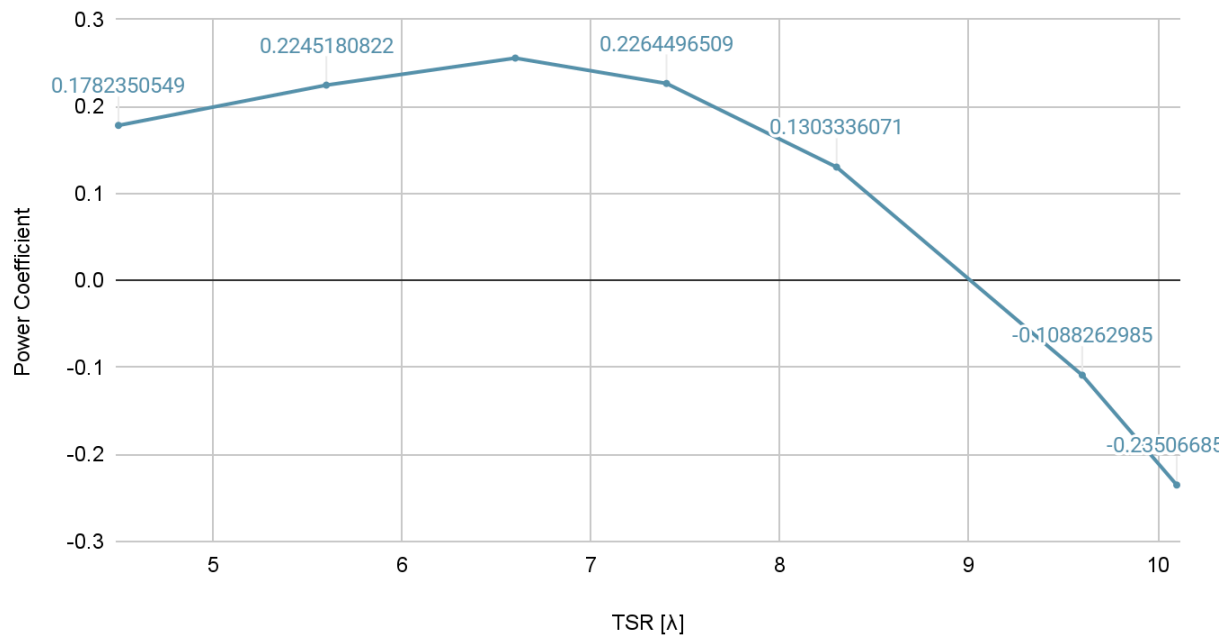
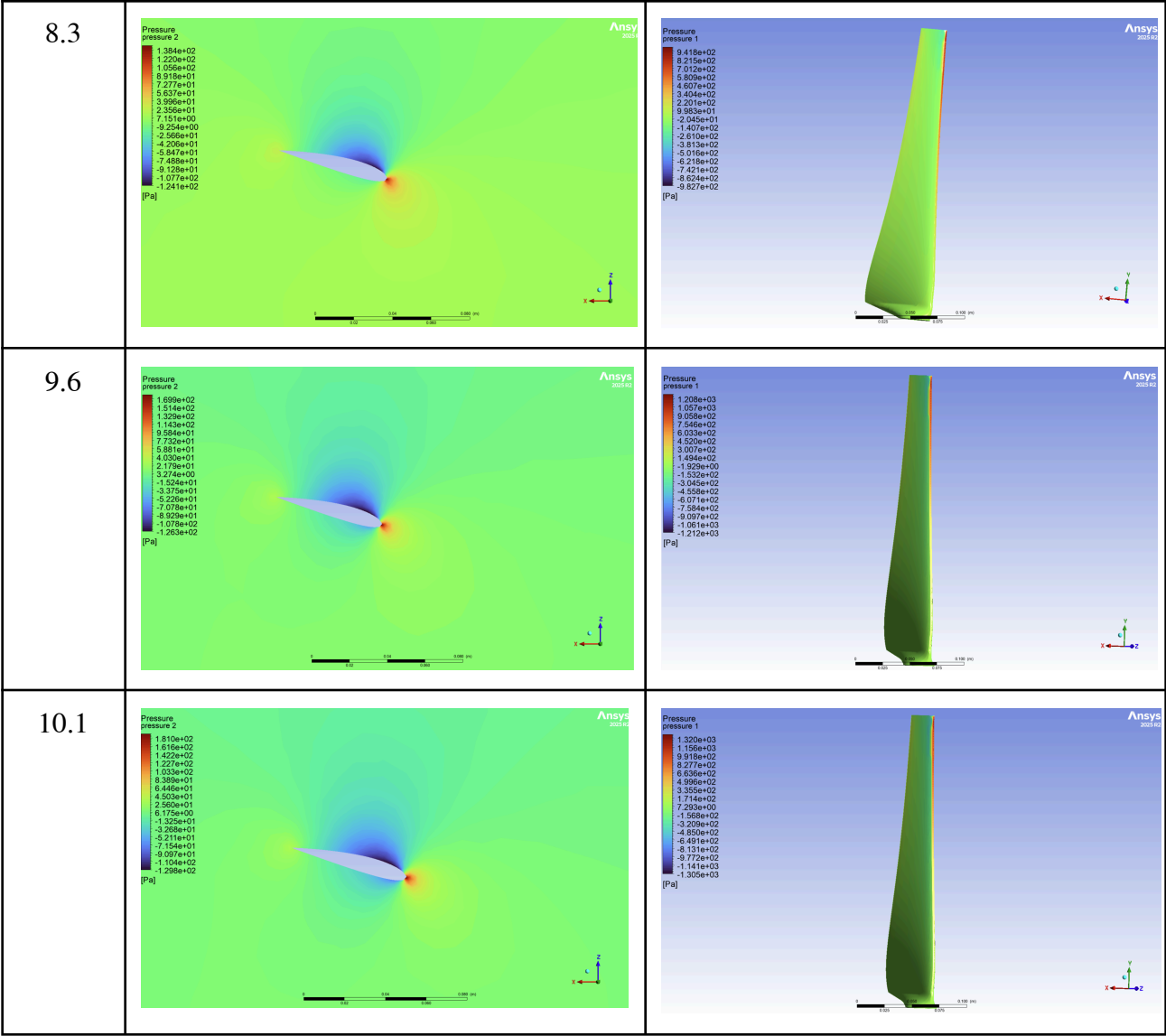


Figure 6: C_p vs λ from Ansys Fluent Data for U=5.5 m/s, $\theta_p = 0^\circ$

Pressure Distributions:

TSR λ_r	XZ Plane Pressure Distribution	Pressure Distribution Along Blade Surface
4.5	Pressure pressure 2 5.105e+01 3.729e+01 2.354e+01 9.785e+00 -3.969e+00 -1.772e+01 -3.144e+01 -4.523e+01 -5.898e+01 -7.274e+01 -8.649e+01 -1.002e+02 -1.140e+02 -1.278e+02 -1.415e+02 -1.553e+02 -1.690e+02 [Pa]	Pressure pressure 1 3.339e+02 2.833e+02 2.527e+02 2.121e+02 1.716e+02 1.310e+02 9.041e+01 4.983e+01 9.249e+00 -3.133e+01 -7.191e+01 -1.125e+02 -1.531e+02 -1.936e+02 -2.342e+02 -2.748e+02 -3.1154e+02 [Pa]
5.6	Pressure pressure 2 6.877e+01 5.705e+01 4.434e+01 3.163e+01 1.891e+01 6.202e+00 -6.511e+00 -1.922e+01 -3.194e+01 -4.465e+01 -5.736e+01 -7.008e+01 -8.279e+01 -9.550e+01 -1.082e+02 -1.209e+02 -1.336e+02 [Pa]	Pressure pressure 1 4.846e+02 4.259e+02 3.654e+02 3.058e+02 2.462e+02 1.866e+02 1.270e+02 6.746e+01 7.874e+00 -5.171e+01 -1.113e+02 -1.709e+02 -2.305e+02 -2.901e+02 -3.497e+02 -4.092e+02 -4.686e+02 [Pa]
6.6	Pressure pressure 2 8.701e+01 7.346e+01 5.991e+01 4.637e+01 3.282e+01 1.927e+01 5.723e+00 -2.137e+01 -3.492e+01 -4.847e+01 -6.202e+01 -7.556e+01 -8.911e+01 -1.027e+02 -1.162e+02 -1.296e+02 [Pa]	Pressure pressure 1 6.313e+02 5.496e+02 4.678e+02 3.861e+02 3.044e+02 2.227e+02 1.409e+02 5.921e+01 -2.251e+01 -1.042e+02 -1.890e+02 -2.677e+02 -3.494e+02 -4.311e+02 -5.129e+02 -5.946e+02 -6.763e+02 [Pa]
7.4	Pressure pressure 2 1.124e+02 9.754e+01 8.270e+01 6.787e+01 5.303e+01 3.819e+01 2.335e+01 8.512e+00 6.325e+00 -2.116e+01 -3.600e+01 -5.084e+01 -6.568e+01 -8.052e+01 -9.536e+01 -1.102e+02 -1.250e+02 [Pa]	Pressure pressure 1 7.740e+02 6.739e+02 5.739e+02 4.738e+02 3.738e+02 2.737e+02 1.737e+02 7.363e+01 -2.642e+01 -1.265e+02 -2.265e+02 -3.266e+02 -4.266e+02 -5.267e+02 -6.267e+02 -7.268e+02 -8.268e+02 [Pa]



Simulation Results vs Wind Tunnel Experimental Data

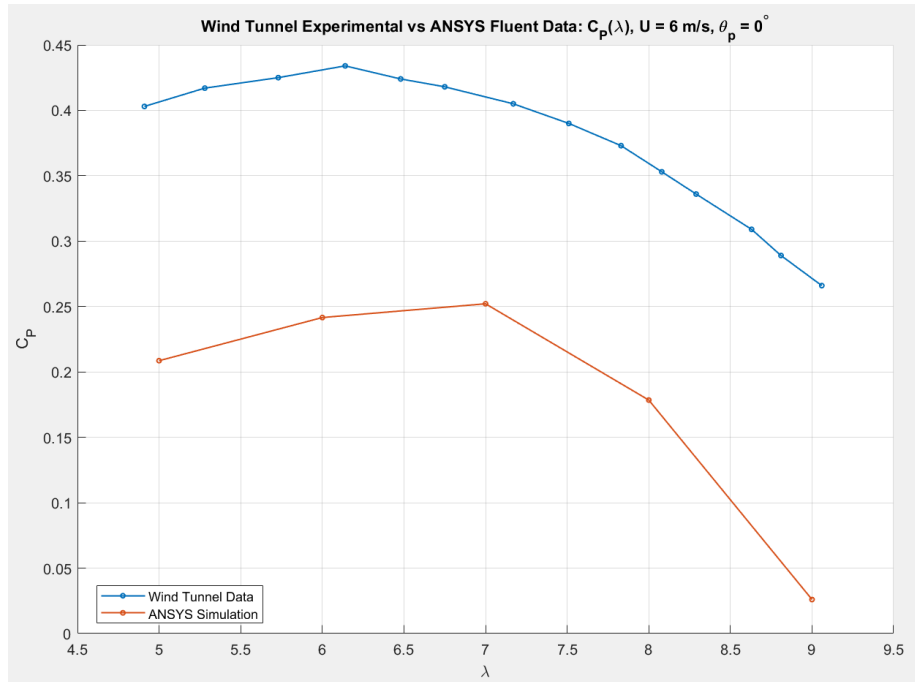


Figure 7: C_p vs λ of NACA 2414 Experimental vs Simulation Data for $U=6 \text{ m/s}$, $\theta_p = 0^\circ$

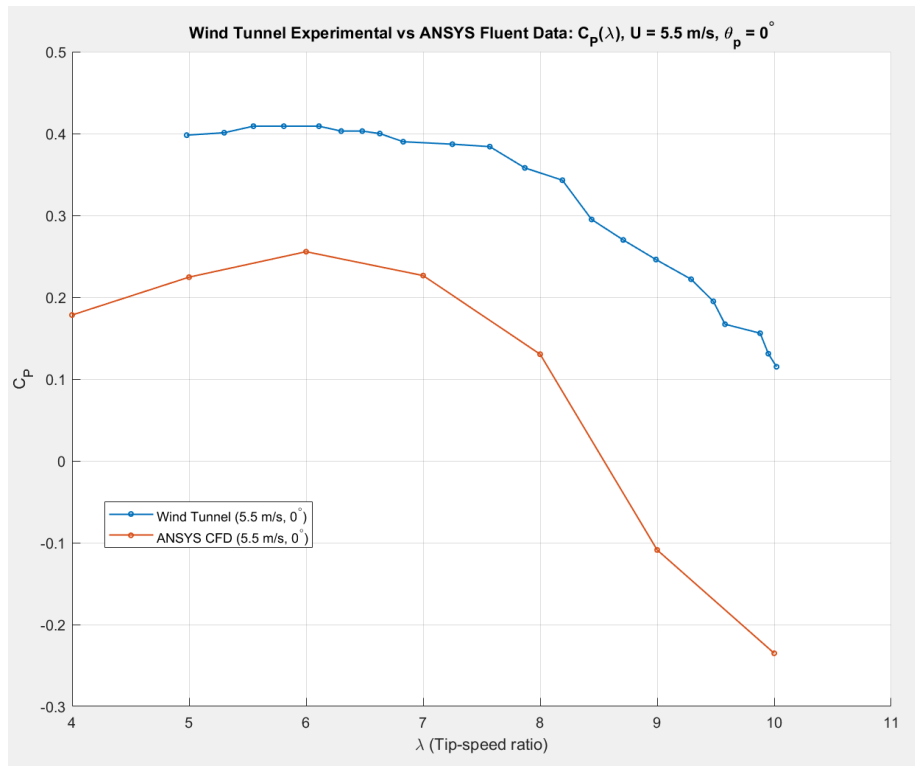


Figure 8: C_p vs λ of NACA 2414 Experimental vs Simulation Data for $U=5.5 \text{ m/s}$, $\theta_p = 0^\circ$

Discussion: Experimental vs Simulation Data for $U=6$ m/s & $U=5.5$ m/s

The comparison between the wind tunnel power-coefficient curves and the ANSYS Fluent predictions at 5.5 m/s and 6 m/s, both at 0° pitch, reveals a consistent under-prediction of C_p by the CFD model across nearly all tip-speed ratios. In the experiments, peak C_p values occur near $\lambda \approx 5.5 - 6.5$ with $C_p \approx 0.40 - 0.43$, whereas the ANSYS simulations peak closer to $C_p \approx 0.24 - 0.26$. The most probable cause of this discrepancy is the extremely coarse CFD mesh ($\sim 60,000$ cells), which is far below typical requirements for resolving boundary-layer physics on low-Re ($Re \approx 50,000$) rotating blades. A coarse mesh cannot capture laminar separation bubbles, transitional flow behavior, or pressure-gradient-driven lift development, all of which dominate small-scale turbine aerodynamics. The result is an over-prediction of drag and an under-prediction of lift, which directly suppresses C_p .

Another notable issue is that several ANSYS cases produced negative torque and negative C_p at the highest tip-speed ratios. Physically, a wind turbine operating in normal inflow cannot produce negative power - the rotor would have to be driven like a fan rather than extracting energy from the wind. This means the negative values are numerical artifacts, not real aerodynamic behavior. These unrealistic results are almost certainly due to numerical instability combined with inadequate resolution at high rotational speeds: coarse surface meshes, low wall-normal refinement, and insufficient wake resolution can cause the solver to mispredict pressure differences between the suction and pressure sides of the blade. Additionally, re-initializing the flow field for each TSR removes the accumulated wake structure from the previous run, making high-TSR cases particularly sensitive to initial imbalances. When the solver fails to converge properly, the predicted surface forces can fluctuate enough to flip the sign of the computed torque.

Additional numerical sources of error include:

1. Reliance on a simple turbulence model (likely $k - \omega$ or SST) that struggles with transitional flows at $Re \approx 50,000$.
2. Re-initializing the flow field for each TSR rather than using a converged solution as the initial condition.
3. The rigid-body rotating reference frame assumption, which may not perfectly represent unsteady wake development.
4. Simplified blade surface definitions compared to the real 3-D printed blades.

Some discrepancy may also exist in the experimental wind-tunnel data, since small wind turbines at low Reynolds numbers are highly sensitive to small geometric imperfections, hub interference, minor twist errors, and reduced inflow uniformity. Even so, the wind-tunnel curves

are smooth, repeatable, and consistent with known performance trends for 2414-based rotors—meaning they serve as a reliable benchmark.

Despite quantitative differences, these comparisons are extremely useful. First, they validate the general shape and location of the performance peak, showing that the CFD captures qualitative behavior even with coarse discretization. Second, the consistent under-prediction confirms that mesh refinement, transition modeling, and improved boundary-layer resolution would meaningfully increase simulation accuracy. These findings allow us to define a realistic performance envelope against which the BEM-optimized geometry can be evaluated, and they highlight the importance of mesh convergence and laminar-turbulent transition modeling for accurate small-scale turbine CFD.

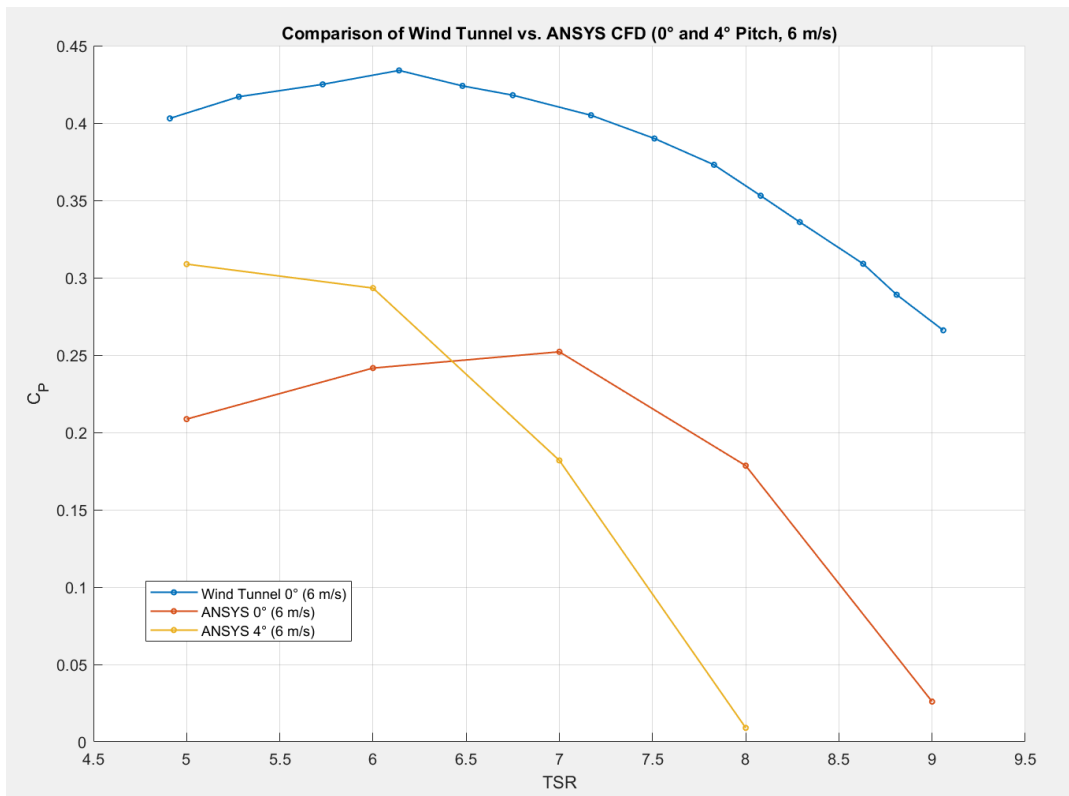


Figure 9: C_p vs λ of NACA 2414 Experimental vs Simulation Data for $U=6$ m/s, $\theta_p = 0^\circ$ & 4°

Discussion: Experimental vs Simulation Data for $U=6$ m/s, Multiple Pitch Angles

The comparison of the wind-tunnel data with the ANSYS CFD results for both 0° and 4° pitch at 6 m/s highlights several important aerodynamic trends and provides insight into the sensitivity of the small rotor to pitch-angle changes. First, the CFD curves remain consistently below the wind-tunnel measurements, but more importantly, the relative locations of their peaks mimic the expected physical behavior: increasing the blade pitch from 0° to 4° decreases the aerodynamic efficiency and shifts the effective operating λ range downward. This is exactly what

should occur - positive pitch reduces the effective angle of attack, weakens lift generation, and therefore lowers torque and C_p . The CFD also captures the sharp drop off at higher TSRs, where stall and reduced relative inflow angles limit lift, although the decline occurs more abruptly in the simulation than in the experiment, again consistent with a coarse mesh or insufficient resolution near the leading edge. The plot is useful because it shows that, despite underpredicting absolute C_p , the CFD model correctly captures the qualitative influence of pitch angle. This means the simulation can still be used as a reliable comparative tool when evaluating how design modifications (e.g., twist, chord, or pitch changes) affect performance trends - even if the absolute values differ from experiment. This validates using CFD as a support tool in the optimization workflow, while also reinforcing the need for finer meshing and improved turbulence modeling to match experimental magnitudes.

Limitations + Future CFD Improvements

Although the CFD simulations provided useful qualitative insight into how pitch angle and TSR influence turbine performance, several modeling limitations restricted the accuracy of the predicted power coefficients. The most significant constraint was the relatively coarse mesh, which limited the model's ability to resolve the thin low-Reynolds-number boundary layers typical of the NACA 2414 airfoil. This can cause premature separation, inaccurate lift and drag prediction, and exaggerated performance degradation at higher TSRs. Additionally, because each new TSR or pitch case required re-meshing and re-initializing the flow field, small inconsistencies were introduced between simulations, reducing the reliability of direct comparisons. The turbulence modeling approach (steady RANS with a basic two-equation closure) also lacks the ability to capture transitional behavior and unsteady vortex structures that strongly affect small-scale rotor aerodynamics. Simplifications in the domain size, rotating frame setup, and treatment of the wake further contributed to discrepancies, especially near peak loading.

Future CFD work can significantly improve accuracy by increasing mesh resolution, adding viscous-layer inflation, and performing mesh-independence studies to ensure that aerodynamics are grid-converged. Implementing a transition-sensitive turbulence model (e.g., SST with intermittency control) would better capture laminar-to-turbulent behavior at $Re \approx 50,000$. Using a consistent mesh with automated pitch adjustments, or a parametric dynamic mesh, would reduce geometric variability across cases. Initializing each TSR case from the previous converged solution rather than restarting from scratch would stabilize convergence and reduce numerical scatter. Finally, expanding the far-field boundaries and refining the wake region would allow more realistic modeling of induction and swirl. These improvements would enhance

quantitative accuracy and make CFD a more dependable verification tool alongside wind-tunnel tests and BEM analysis.

Blade Design Optimization

BEM Model of Standard Blade

The standard-blade model, as seen in Appendix A, uses Blade Element Momentum (BEM) theory to predict the power coefficient $C_p(\lambda)$ of the given lab turbine blade geometry across a range of tip-speed ratios. The geometry is not designed in this script - instead, it directly uses the measured standard blade distributions of radius stations, chord, and local pitch: `r_local` holds the radial locations of blade sections, `chord_vec` stores the corresponding chord lengths, and `theta_p_dist` contains the local pitch angle in degrees from hub to tip. At each radial station, the model combines these geometric inputs with NACA 2414 airfoil polar data (lift and drag coefficients as functions of angle of attack) read from `xf-n2414-il-50000.csv`. For a given tip-speed ratio λ , the code computes the rotor angular speed $\Omega = \lambda U/R$ and then steps through each blade element, solving for the axial and tangential induction factors a and a' by iterating the coupled momentum and blade-element equations.

Physically, the script follows the full BEM Method 2 formulation from Manwell et al. At each radial section, it computes the local inflow angle ϕ from the axial and tangential components of the relative velocity and then calculates the local angle of attack as $\alpha = \phi - \theta_p(r)$. This angle is clamped to the airfoil polar range and used to interpolate the local lift and drag coefficients C_l and C_d . These are then resolved into normal and tangential force coefficients relative to the rotor plane, C_{fn} and C_{ft} , which drives both the axial (thrust) and tangential (torque) balance. The model applies Prandtl's tip-loss factor to correct for the finite number of blades and evaluates a local thrust coefficient to decide whether to use the standard momentum expression for a or a high-thrust/Glauert-style correction when the loading becomes large. Within each element, the axial and tangential induction factors are iteratively updated until they converge below a specified tolerance, using a small relaxation factor to improve numerical stability. Finally, the script uses a discrete form of Manwell's Eq. 3.136 to accumulate the contribution of each element to the overall power coefficient, effectively integrating the elemental torque and normal-force effects along the blade span. Repeating this process for all tip-speed ratios in `TSR_range` produces a $C_p(\lambda)$ curve for the standard blade, and the script identifies and marks the C_p and corresponding optimal λ_{opt} directly on the plot. This gives a consistent BEM-based performance baseline that can be compared against both wind-tunnel data and the optimized blade design.

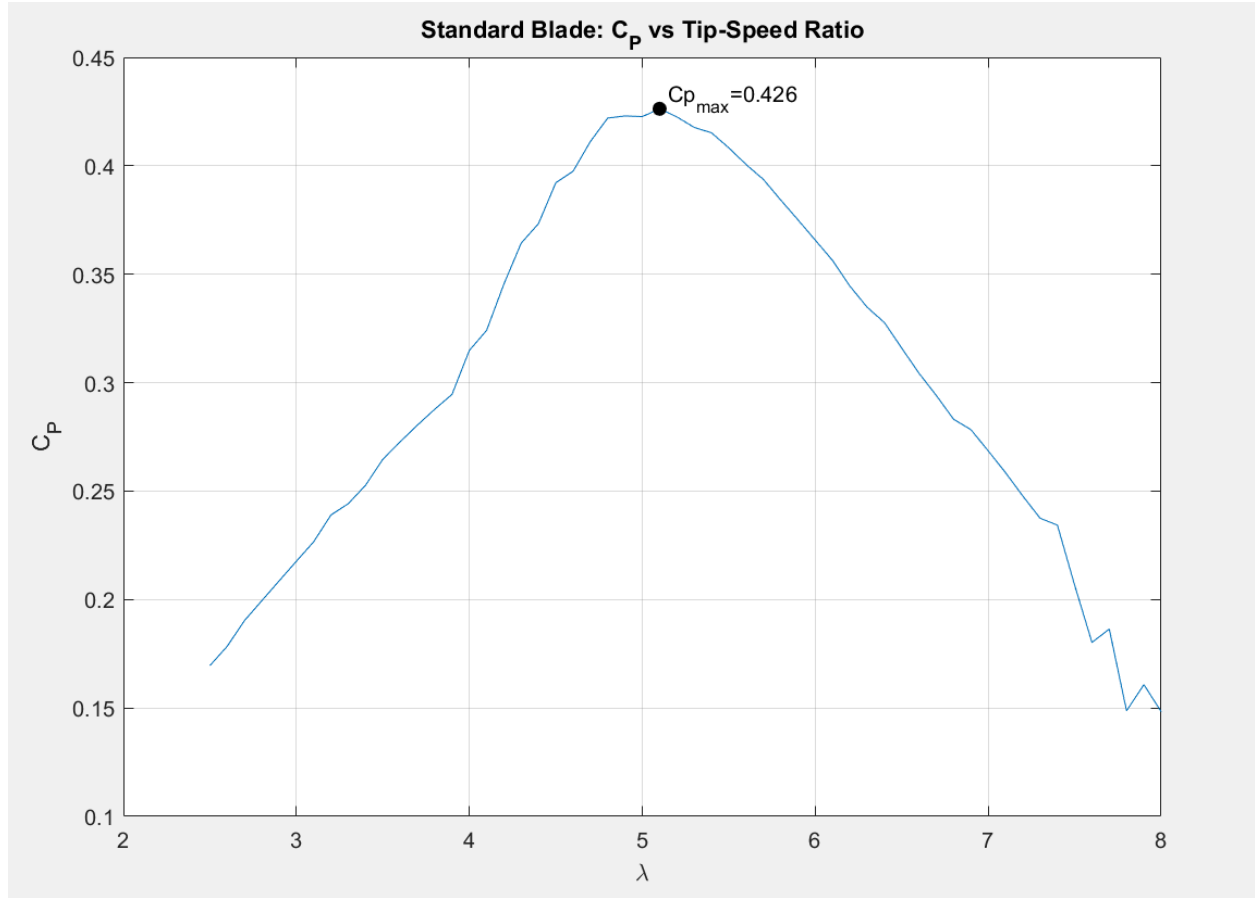


Figure 10: Standard NACA 2414 Blade Behavior Modeled by BEM in MATLAB

Comparison Between Standard BEM Model and Wind-Tunnel Results

The performance of the standard laboratory blade predicted by the BEM model aligns closely with the experimental wind-tunnel data. The BEM simulation produced a peak power coefficient of approximately $C_p = 0.426$ at $\lambda = 5.1$, while the wind-tunnel measurements for the same blade at 0° pitch and 6 m/s reported a slightly higher peak of $C_p = 0.434$ occurring near $\lambda = 6.1$. This difference of less than 2% in C_p demonstrates strong agreement between theory and experiment, validating both the aerodynamic model and the input geometry used for the standard blade. The small discrepancy in peak location and magnitude is expected. BEM theory, while reliable, makes several simplifying assumptions, including uniform inflow, axisymmetric loading, and steady flow without turbulence or 3D stall effects. It also uses 2D airfoil polar data at fixed Reynolds numbers, which may not perfectly represent the actual, three-dimensional rotating blade operating in a real wind tunnel where Reynolds number varies along the span. Tip-vortex behavior, root flow losses, and slight deviations in the manufactured blade geometry can also influence the measured torque. On the experimental side, uncertainty in

torque calibration, subtle misalignment of pitch angle, surface roughness on the blade, and blockage or interaction effects inside the tunnel can shift the apparent performance. Together, these factors explain why the wind-tunnel peak appears at a slightly higher TSR and with a marginally higher C_p , while still showing excellent agreement with the theoretical BEM prediction.

BEM Model for Blade Design Optimization

The blade-optimization script uses the full Blade Element Momentum (BEM) Method 2 to redesign the chord and twist distribution of a NACA 2414 wind turbine blade in order to achieve a higher power coefficient than the standard laboratory blade, while maintaining the original rotor radius. The process begins by importing NACA 2414 lift and drag polar data and determining an aerodynamic design point based on the maximum lift-to-drag ratio, which defines the optimal angle of attack used throughout the geometric construction. For each candidate design, the script sweeps a range of design tip-speed ratios from 3.5-5.5 and, following the optimal rotor theory in Manwell et al. (Eqs. 3.112 - 3.113), computes a corresponding chord distribution and twist (pitch) distribution along the blade span using the analytic expression for the ideal relative wind angle $\phi = \frac{2}{3}\arctan(1/\lambda_r)$. These candidate geometries are then evaluated using the iterative BEM Method 2 formulation (Eqs. 3.127 - 3.133), which solves for the axial and tangential induction factors at each of the 100 radial stations by iteratively balancing the blade-element aerodynamic forces with the momentum equations. Each iteration incorporates tip-loss corrections using Prandtl's exponential tip-loss factor, and automatically switches to the Glauert high-thrust correction when the local thrust coefficient exceeds 0.96, ensuring stability and physical accuracy under heavy loading. At each radial element, the airfoil polars are re-interpolated based on the instantaneous angle of attack computed from the converged flow angle. After convergence of the induction factors, the script computes the elemental contribution to the rotor power using the discrete form of Manwell's power equation (Eq. 3.136), integrating these contributions along the span to obtain a full $C_p(\lambda)$ curve for each geometry. The algorithm selects the best design as the one producing the highest maximum power coefficient across the TSR sweep, then outputs its optimized chord and twist distributions as well as the optimal TSR at which peak performance occurs. Finally, the script automatically calls the standard-blade performance function and generates a comparison plot, overlaying the standard blade's $C_p(\lambda)$ curve with that of the optimized blade, including markers for both peak values. This end-to-end process provides a fully automated BEM-based geometric optimization that directly aligns with the theoretical framework of Chapter 3 in *Wind Energy Explained*, enabling a controlled, physics-based improvement over the standard blade geometry.

Optimized NACA 2414 Blade Using BEM Theory

Using the BEM-based optimization framework described in the prior section, the final blade design that maximizes power coefficient under the constraints of a fixed rotor radius ($R = 0.305$ m), NACA 2414 airfoil, and manufacturable chord/twist profiles is summarized by the chord distribution, pitch distribution, and predicted performance curves shown in Figures 12-13. The optimization algorithm swept a range of design tip-speed ratios $\lambda_{design} = 3.5 - 5.5$ and evaluated the performance of each resulting blade shape using the full BEM Method 2 formulation. The design producing the highest predicted peak power coefficient was obtained at $\lambda = 4.9$, yielding a maximum BEM-predicted $C_p \approx 0.440$, which exceeds the measured C_p of the standard turbine (~ 0.40 - 0.43) while maintaining a manufacturable, monotonically decreasing chord profile and smoothly tapered twist.

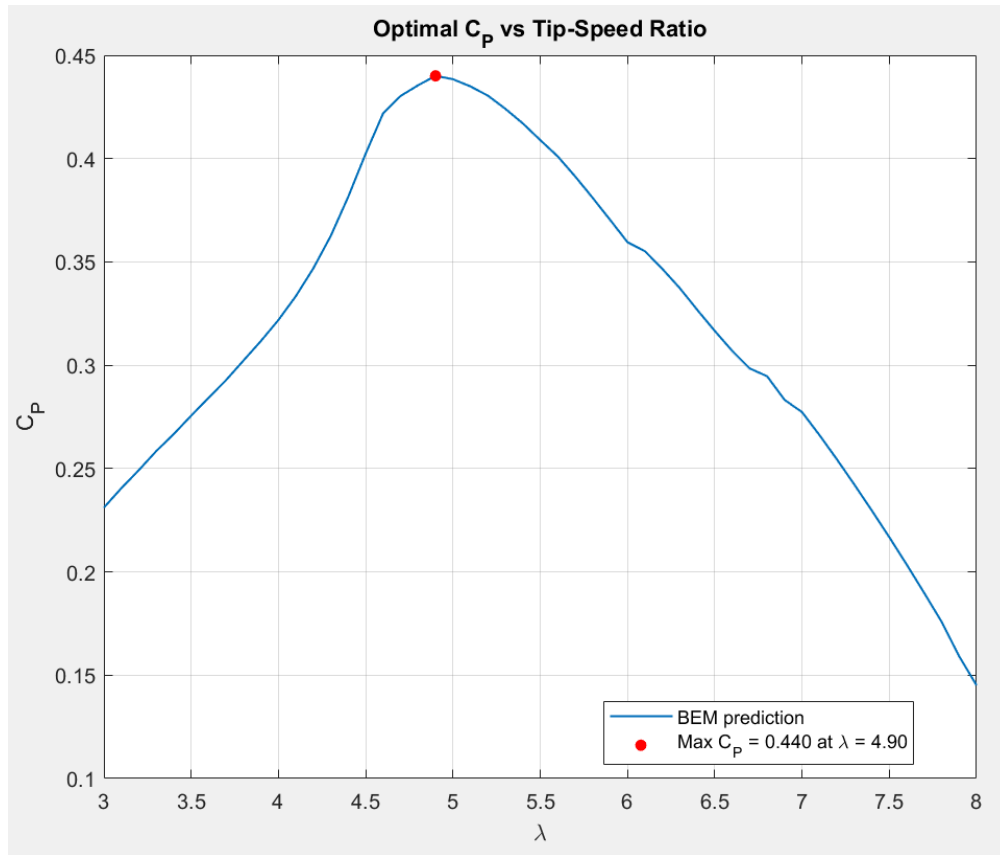


Figure 11: C_p vs λ for Optimized Blade Design

The optimized chord distribution, shown in Figure 12, exhibits the characteristic taper recommended in classical rotor theory: a relatively wide root region followed by a smooth reduction toward the tip. Chord lengths range from approximately 0.06 m near the root down to ~ 0.02 m at the tip, ensuring structural robustness while reducing induced losses. This shape

closely resembles the classical ideal Betz distribution modified for finite-blade effects and low-Reynolds-number operation.

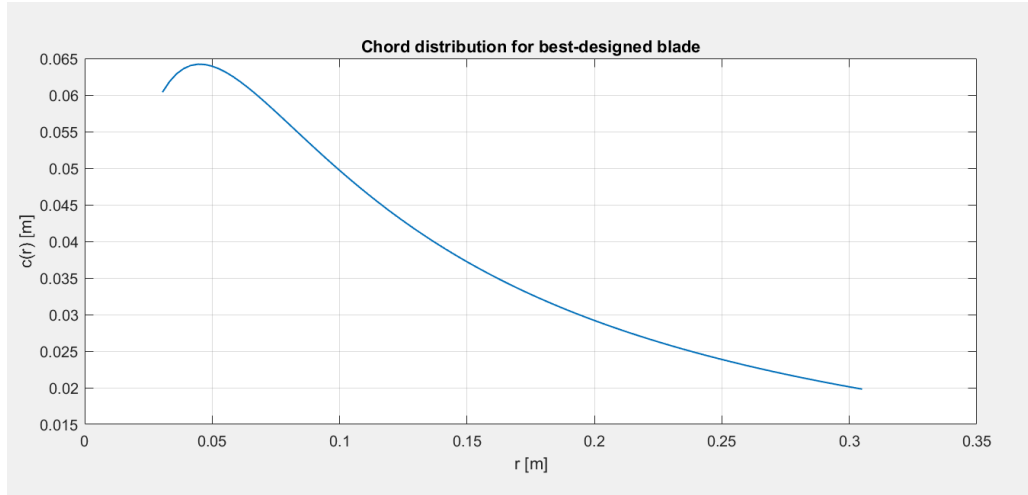


Figure 12: Chord Distribution for Optimized Blade Design

The corresponding twist (pitch) distribution in Figure 13 shows a steep twist near the hub, around 33° at $r \approx 0.03$ m, that gradually reduces toward a near-zero twist at the tip ($\sim -1.3^\circ$ at $r = 0.305$ m). This pattern is physically consistent: a large twist is required near the hub to maintain a favorable angle of attack despite low local relative wind speeds, while the tip experiences high relative velocity and therefore requires minimal twist. The negative pitch near the tip is characteristic of low-Re, small rotors using cambered airfoils and inclusion of wake rotation effects in BEM theory.

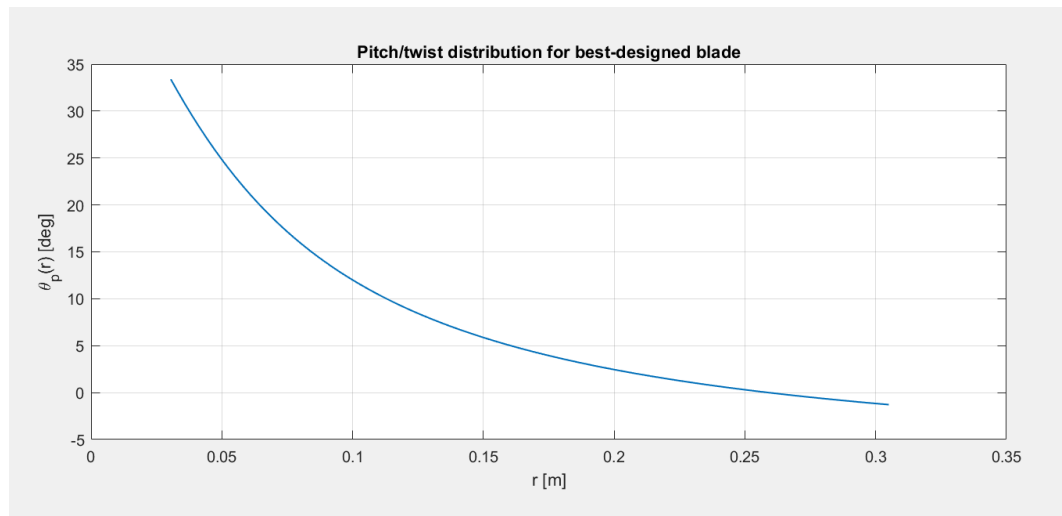


Figure 13: Pitch Distribution for Optimized Blade Design

The final optimized geometry is summarized numerically in Table 1, which lists the radial stations used in the computation along with their corresponding chord and pitch values. These radial stations match those used in the computational model and represent the exact surface parameters required to fabricate or CAD-model the blade.

Optimized Geometry:

Radius (m)	Chord (m)	Pitch (deg)
0.0305	0.060384	33.387
0.061	0.061827	21.06
0.0915	0.052349	13.535
0.122	0.043539	8.8085
0.1525	0.036718	5.6537
0.183	0.031535	3.4264
0.2135	0.027542	1.7803
0.244	0.024401	0.51846
0.2745	0.021877	-0.47759
0.305	0.019813	-1.2828

Table 1: Geometry for Optimized Blade Design

Comparison of Blade Performance: Standard vs. Optimized Model

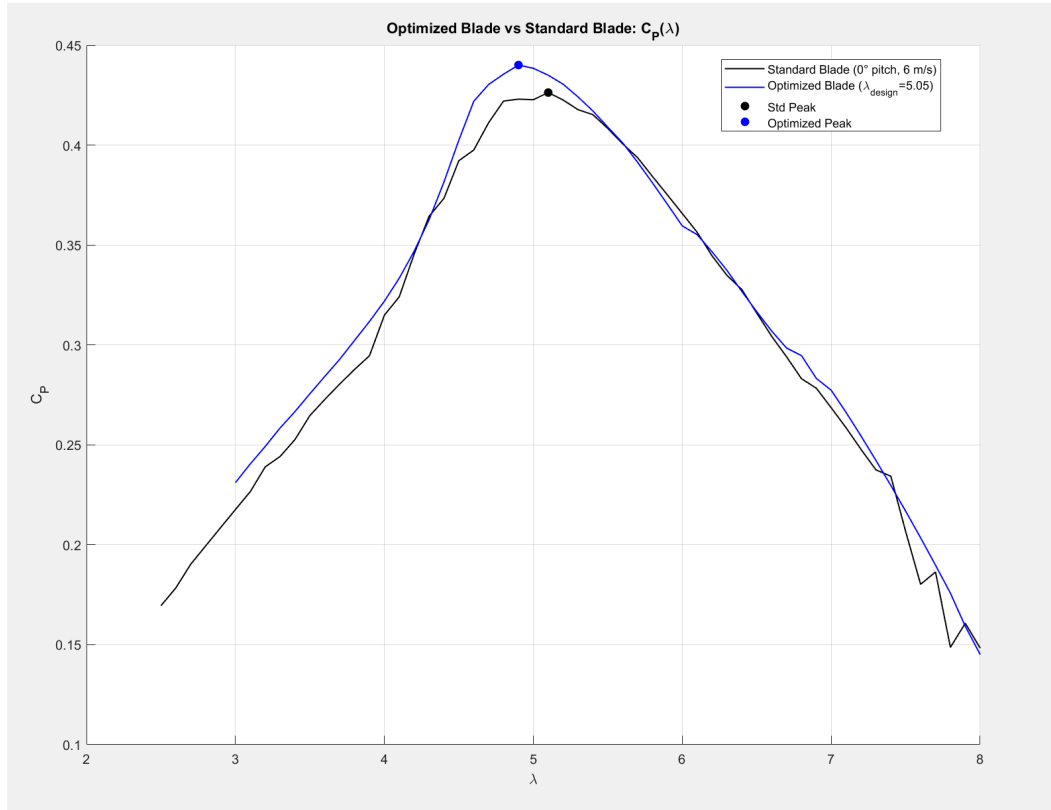


Figure 14: C_p vs λ for Standard vs Optimized Blade Design

This final comparison between the optimized blade and the BEM-modeled standard blade (both using NACA 2414 and $R = 0.305$ m) shows that relatively modest geometric changes can produce a meaningful gain in predicted performance over a broad range of tip-speed ratios. The optimized blade (blue curve) has a slightly higher and sharper peak, reaching $C_p \approx 0.440$ at $\lambda \approx 4.9$, whereas the standard blade (black curve) peaks around $C_p \approx 0.43$ at a slightly higher λ . This 3–4% increase in peak power coefficient is achieved while keeping the same rotor radius and airfoil and only reshaping the chord and twist distributions. At lower TSR (~ 3 – 4) the optimized blade also climbs faster than the standard one, indicating that it loads the inner radii more effectively and reaches useful power levels sooner as the rotor spins up. At high TSR (~ 6.5 – 8) the two curves nearly collapse, suggesting that the optimization did not sacrifice off-design robustness: when the rotor overspeeds, both designs roll off in a very similar way.

When these BEM results are interpreted alongside the wind-tunnel tests at multiple tip-speed ratios and pitch settings, the picture is consistent. Experimentally, the standard blade at 0° and -2° pitch showed broad $C_p(\lambda)$ peaks around $\lambda \approx 5.5 - 6.5$ and $C_p \approx 0.40 - 0.43$. The optimized design simply nudges that performance envelope upward and slightly left, essentially “retuning” the spanwise chord and pitch so that, at a design TSR of about 5, more of

the blade operates near the airfoil's best C_l/C_d while still maintaining reasonable loading at neighboring TSRs. The twist schedule - large pitch near the hub, decaying toward slightly negative pitch at the tip - keeps the local angle of attack closer to the optimum across the radius for each operating condition, which is exactly what the textbook Method-2 BEM design is intended to do.

A particularly important implication of this optimization is the lower optimal TSR. A design TSR of ~ 4.9 is noticeably lower than the TSRs at which the standard blade typically attains its peak performance (~ 5.7 - 6.3). This shift has several meaningful consequences:

A lower optimal TSR reduces blade rotational speed for a given wind speed, which leads directly to lower tip Mach numbers, reduced noise emissions, and reduced aerodynamic loading fluctuations. Lower rotational speeds also mitigate leading-edge erosion, one of the dominant lifetime-limiting mechanisms for small turbines, especially those fabricated from softer materials or operating in dusty or icy environments. Reduced centrifugal loads further decrease structural stresses in the blade, hub, and root attachments, improving fatigue life and reducing the likelihood of delamination or crack growth over long-term use.

On the other hand, a lower design TSR also means the rotor extracts more torque at lower rotational speeds, which can impose a higher static or low-speed startup torque requirement on the drivetrain or generator. If the electrical system is not sized for this shift, it may reduce low-wind-speed performance or cause suboptimal generator loading. Additionally, operating at lower TSRs can increase wake turbulence and slightly enlarge the aerodynamic footprint of the rotor, which might matter in tightly spaced arrays or in classroom/wind-tunnel settings where recirculation effects are present.

Overall, these results provide strong evidence that the optimized blade geometry offers a modest but meaningful increase in aerodynamic efficiency while also delivering several structural and operational advantages tied to its lower design TSR. Together with the wind-tunnel and CFD comparisons, this establishes a realistic and physically justified performance improvement relative to the original blade design, and it does so without sacrificing robustness across a wide operating range.

Conclusions and Future Work

The work completed in this project demonstrates that meaningful aerodynamic improvements can be achieved for the course wind turbine blade using a combination of textbook Blade Element Momentum (BEM) design methods, CFD analyses, and comparisons to experimental wind-tunnel data. By optimizing the chord and twist distributions while maintaining the original radius and NACA 2414 airfoil, the redesigned blade achieved a modest yet significant increase in predicted peak power coefficient - reaching $C_p \approx 0.44$, compared to approximately 0.42-0.43 for the standard blade under identical operating conditions. This improvement was accomplished without increasing geometric complexity, confirming that

classical BEM-based optimization still provides robust performance gains for small-scale rotors at low Reynolds numbers. Furthermore, the optimized blade features a lower design TSR, which not only contributes to improved aerodynamic loading but also implies reduced noise, lower leading-edge erosion rates, and decreased centrifugal stresses. Taken together, the results validate the potential for redesigning existing small turbines to achieve higher efficiency while extending operational lifespan.

However, the comparisons between CFD and wind-tunnel data also highlighted important limitations. The ANSYS Fluent simulations, conducted on a coarse mesh of ~60,000 cells and using a simplified turbulence model, consistently under-predicted performance relative to experiments. This suggests that more accurate CFD simulations will require substantially finer near-wall resolution, transition-sensitive turbulence models, and careful meshing around the blade leading edge to properly resolve laminar separation bubbles. Additionally, negative predicted torque and C_p at high TSR indicate numerical instability or insufficient mesh refinement, emphasizing that future CFD work should include systematic mesh independence studies, improved initialization strategies, and rotating-frame sensitivity analyses.

Looking forward, several clear paths exist for expanding and deepening this work. Experimentally, repeating the wind-tunnel tests across a wider range of wind speeds, pitch settings, and blade geometries would allow for thorough validation of both the optimized design and the BEM predictions. It would also be valuable to acquire higher-fidelity airfoil polar data at multiple Reynolds numbers, as small turbines are highly Reynolds-dependent and sensitive to surface roughness. These expanded protocols would produce a significantly richer dataset, allowing the optimized blade to be validated under the same breadth of conditions expected in real operation. They would also serve as a benchmark for improving CFD and BEM correlation.

Beyond aerodynamic validation, the redesigned blade should undergo structural and dynamic evaluations. A finite element stress analysis could verify that the modified chord distribution does not induce excessive root bending stress or unfavorable stress concentrations. Fatigue-life analysis would characterize durability under cyclic aerodynamic loading. Vibration testing, both analytical modal analysis and experimental modal surveys, would ensure that the new mass and stiffness distributions do not place resonant frequencies near operating rotational speeds, preventing harmful amplification.

Another important extension would be to explore optimization under variable wind conditions. The current design is optimized for a single TSR and a fixed free-stream wind speed, whereas real wind turbines operate across a distribution of wind velocities. A more advanced optimization could incorporate a Weibull wind-speed distribution or perform multi-objective optimization across a range of TSRs to produce a blade with robust, all-conditions performance.

This could also involve exploring passive or active pitch-control strategies, variable-speed generator matching, or lifting-line/BEM-CFD hybrid optimization approaches.

Overall, this project lays a strong foundation: a validated BEM optimization framework, reasonable CFD and experimental benchmarks, and an initial demonstration that performance gains are achievable. Future work can expand this foundation toward a full aero-structural design study, a more sophisticated CFD validation campaign, and a holistic optimization that reflects the true operating environment of small horizontal-axis wind turbines.

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Appendix A: MATLAB Programs

Standard blade cp curve rag323.m

```
function[TSR_range,CP_std]=standard_blade_cp_curve_rag323()
data=readtable('xf-n2414-il-50000.csv'); % alpha [deg], Cl, Cd
alpha_val=data(:,1);
lift_c=data(:,2);
drag_c=data(:,3);
% parameters
U_wind=6;
radius=0.305;
B=3;
% radial points for standard blade (m)
r_local=[0.025,0.050,0.075,0.100,0.125,0.150,0.175,0.200,0.225,0.250,0.2648];
```

```

% local pitch distribution (deg)
theta_p_dist=[22.27,16.04,11.82,8.67,6.45,4.57,3.16,2.13,1.27,0.46,0.00];
% chord distribution (m)
chord_vec=[0.0686,0.061180471,0.053228,0.046430701,0.040959126,0.036415656,0.03
2649655,0.029620432,0.027106641,0.024900803,0.0237];
N=numel(r_local); %number of blade elements
TSR_range=2.5:0.1:8; %TSR sweep
CP_std=zeros(size(TSR_range));
for w=1:length(TSR_range)
    lambda=TSR_range(w); %current TSR
    omega=lambda*U_wind/radius; %angular velocity
    cp_sum=0;
    for n=1:N
        r_n=r_local(n);
        c_n=chord_vec(n);
        lambda_r=omega*r_n/U_wind;
        sigma_n=B*c_n/(2*pi*r_n);
        %initialize induction factors
        a_i=0.3;
        ap_i=0.0;
        tol=1e-5;

        %iterative solution for a_i and ap_i
        for iter=1:100
            %axial and tangential velocity components
            U_ax=U_wind*(1-a_i);
            U_tan=omega*r_n*(1+ap_i);
            phi=atan2(U_ax,U_tan);%inflow angle
            phi_deg=rad2deg(phi);
            alpha_deg=phi_deg-theta_p_dist(n); %local AoA
            alpha_clamped=max(min(alpha_deg,max(alpha_val)),min(alpha_val));
            Cl=interp1(alpha_val,lift_c,alpha_clamped,'linear');
            Cd=interp1(alpha_val,drag_c,alpha_clamped,'linear');

            %force coefficients in normal/tangential directions
            C_fn=Cl*cos(phi)+Cd*sin(phi);
            C_ft=Cl*sin(phi)-Cd*cos(phi);
            %prandtl tip loss correction
            sph_i=max(sin(phi),1e-6);
            F=(2/pi)*acos(exp((-0.5*B*(1-r_n/radius))/(r_n/radius)*sph_i));
            F=max(F,1e-3);
            %local thrust coefficient
            C_Tloc=sigma_n*(1-a_i)^2*C_fn/max(sph_i^2,1e-8);
            %update a_i
            if C_Tloc<0.96
                a_new=1/(1+(4*F*sph_i^2)/(sigma_n*C_fn));
            else
                a_new=(1/F)*(0.143+sqrt(0.0203-0.6427*(0.889-C_Tloc)));
            end
        end
    end
end

```

```

end
ap_new=1/((4*F*sphi*cos(phi))/(sigma_n*C_ft+1e-8)-1);
%convergence check
if abs(a_new-a_i)<tol && abs(ap_new-ap_i)<tol
    a_i=a_new;
    ap_i=ap_new;
    break;
end
%relaxation to ensure stable iteration
damp=0.5;
a_i=damp*a_new+(1-damp)*a_i;
ap_i=damp*ap_new+(1-damp)*ap_i;
end
cotphi=cos(phi)/max(sin(phi),1e-6);
add_term=F*sphi^2*(cos(phi)-lambda_r*sphi)*(sphi+lambda_r*cos(phi))*...
    (1-(Cd/Cl)*cotphi)*lambda_r^2;
cp_sum=cp_sum+add_term; %add up Cp contributions
end
CP_std(w)=(8/(lambda*N))*cp_sum; %final Cp for this TSR
end
%plot results
figure;
plot(TSR_range,CP_std); hold on;
grid on;
xlabel('\lambda'); ylabel('C_P');
title('Standard Blade: C_P vs Tip-Speed Ratio');
[CP_max_std,idx_max]=max(CP_std);
lambda_peak_std=TSR_range(idx_max);
% Mark the peak Cp
plot(lambda_peak_std,CP_max_std,'ko','MarkerFaceColor','k');
% Add labels near the markers
text(lambda_peak_std+0.05,CP_max_std+0.005,sprintf('Cp_{max}=%.3f',CP_max_std))
;
text(lambda_peak_std+0.05,0.02,sprintf('\lambda_{opt}=%.2f',lambda_peak_std));
fprintf('STANDARD blade: max Cp = %.4f at lambda =
%.2f\n',CP_max_std,lambda_peak_std);
end

```

BEM DesignOpt RiyaGuttigoli rag323.m

```

clear; clc; close all;
%read airfoil data
data = readtable('xf-n2414-il-50000.csv');
alpha_val=data(:,1);
lift_c=data(:,2);% Cl
drag_c=data(:,3);% Cd
% design angle of attack using Cl/Cd
ratios=lift_c ./ drag_c;

```

```

    [~, max_index]=max(ratios);
    opt_alpha=alpha_val(max_index);
    Cl_best= lift_c(max_index);
    Cd_best=drag_c(max_index);
%blade parameteres
    radius=0.305;
    B= 3;
    N=100;
    U_wind=6; % design wind speed
% discretize blade elements
    r_root= 0.1*radius; %avoids irregularities at hub
    r_local=linspace(r_root, radius, N);
    dr=r_local(2)-r_local(1);
% TSR range for performance sweep
    TSR_range=3:0.1:8;
% sweep of design TSR values to try
    lambda_design_list = 3.5:0.05:5.5;
% trackers for best design parameters
    bestCp= -inf;
    best_lambda_design=NaN;
    best_lambda_opt= NaN;
    best_CP_curve =[];
    best_chord_vec=[];
    best_theta_p_deg =[];
    best_r_local =r_local;
for k = 1:length(lambda_design_list)
    %optimize blade using textbook procedure for this design TSR
    lambda_design=lambda_design_list(k);%chosen design TSR
    lambda_r_design=lambda_design*(r_local / radius);
    phi_design=(2/3)*atan(1 ./ lambda_r_design);
% chord distribution - eq. 3.112
    chord_vec=(8*pi*r_local).*(1-cos(phi_design))/(B*Cl_best);
% pitch/twist distribution - eq. 3.113
    theta_p_dist=phi_design-deg2rad(opt_alpha);% rad
%Method 2 BEM to get Cp as function of lambda
    CP=zeros(size(TSR_range));
for w=1:length(TSR_range)
    lambda=TSR_range(w);
    omega=lambda*(U_wind/radius);
    % Start summing Cp - eq. 3.136
    cp_sum=0;
    %Loop through blade elements
    for n=1:N
        r_n=r_local(n);
        lambda_r=omega*r_n/U_wind;
        %local chord and solidity
        chord=chord_vec(n);
        solidity=B*chord/(2*pi*r_n);

```

```

% initial guesses for induction factors
a_i = 0.3;
ap_i = 0;
iterations=100;
tolerance=1e-5;
for it=1:iterations
    % flow angle - Eq 3.127
    phi=atan(U_wind*(1-a_i)/(omega*r_n*(1+ap_i)));
    phi_deg=rad2deg(phi);

    % operating angle of attack
    alpha =phi_deg-rad2deg(theta_p_dist(n));

    % clamp alpha to data range
    alpha_clamped=max(min(alpha, max(alpha_val)),min(alpha_val));

    % airfoil coefficients
    Cl= interp1(alpha_val, lift_c, alpha_clamped, "linear");
    Cd=interp1(alpha_val, drag_c, alpha_clamped, "linear");

    % normal and tangential force coeffs
    C_fn=Cl*cos(phi)+Cd*sin(phi);
    C_ft=Cl*sin(phi)-Cd*cos(phi);

    % tip loss factor eq 3.128
    sphi=max(sin(phi),1e-6);
    F=(2/pi)*acos(exp( (-0.5*B*(1-r_n/radius))
/((r_n/radius)*sphi)));

    % local thrust coeff eq 3.130
    c_t=solidity*(1-a_i)^2*(Cl*cos(phi) +Cd*sin(phi))/(sphi^2);

    % update a and a'
    if c_t<0.96
        a_new =1/(1 + (4*F*sphi^2)/(solidity*C_fn));
        ap_new=1/((4*F*sphi*cos(phi))/(solidity*C_ft)-1);
    else
        a_new=(1/F)*(0.143+sqrt(0.0203-0.6427*(0.889-c_t)));
        ap_new=1/((4*F*cos(phi))/(solidity*C_ft) - 1);
    end

    % convergence with damping
    if abs(a_new-a_i)<tolerance && abs(ap_new-ap_i)<tolerance
        a_i =a_new;
        ap_i=ap_new;
        break;
    end

    damp=0.3;
    a_i=damp*a_new + (1-damp)*a_i;

```

```

        ap_i=damp*ap_new+(1-damp)*ap_i;
    end

    % Eq. 3.136
    cotphi=cos(phi)/max(sin(phi),1e-6);
    add_on=F*sphi^2 *(cos(phi)-lambda_r*sphi) *(sphi+
lambda_r*cos(phi))*(1 - (Cd/C1)*cotphi)*lambda_r^2;
    cp_sum=cp_sum + add_on;
end
% final Cp for this lambda eq 3.136
CP(w)=(8/(lambda*N))*cp_sum;
end
%find and print best CP and TSR for this design TSR
[CP_max, idx_max]=max(CP);
lambda_opt=TSR_range(idx_max);
fprintf('lambda_design = %.2f --> Max Cp = %.3f at TSR =
%.2f\n',lambda_design,CP_max,lambda_opt);
% update global best design based on Cp_max
if CP_max > bestCp
    bestCp=CP_max;
    best_lambda_design=lambda_design;
    best_lambda_opt=lambda_opt;
    best_CP_curve=CP;
    best_chord_vec=chord_vec;
    best_theta_p_deg=rad2deg(theta_p_dist);
end
end
% summary of best design over all design lambdas
fprintf('\nBest Design over lambda_design Sweep:\n');
fprintf('lambda_design= %.2f\n', best_lambda_design);
fprintf('Max Cp= %.3f at TSR = %.2f\n', bestCp, best_lambda_opt);
%print geometry table for best design
% choose 10 evenly spaced regions along the blade
Nout=10;
idx_out=round(linspace(1,length(best_r_local),Nout));
radius_opt=best_r_local(idx_out).';
chord_opt=best_chord_vec(idx_out).';
pitch_opt=best_theta_p_deg(idx_out).';
geom_table=table(radius_opt,chord_opt,pitch_opt,'VariableNames',{'Radius
(m)','Chord (m)','Pitch (deg)'});
disp('Optimized Geometry:');
disp(geom_table);
%plot geometry c(r) and pitch(r) for best design
figure;
subplot(2,1,1);
plot(best_r_local, best_chord_vec, '-', 'LineWidth',1);
grid on;
xlabel('r [m]');

```

```

ylabel('c(r) [m]');
title('Chord distribution for best-designed blade');
subplot(2,1,2);
plot(best_r_local, best_theta_p_deg, '-', 'LineWidth',1);
grid on;
xlabel('r [m]');
ylabel('\theta_p(r) [deg]');
title('Pitch/twist distribution for best-designed blade');
%CP lambda plot with marked peak for best design
figure;
plot(TSR_range, best_CP_curve, '-', 'LineWidth',1); hold on;
plot(best_lambda_opt, bestCp, 'ro', 'MarkerFaceColor','r', 'MarkerSize',5);
grid on;
xlabel('\lambda');
ylabel('C_P');
title(sprintf('Optimal C_P vs Tip-Speed Ratio'));
legend('BEM prediction', sprintf('Max C_P = %.3f at \lambda = %.2f', bestCp,
best_lambda_opt), 'Location', 'best');
%call standard blade function and map against
[TSR_std, CP_std]=standard_blade_cp_curve_rag323();
% Plot comparison
figure; hold on;
plot(TSR_std, CP_std, 'k', 'LineWidth', 1);
plot(TSR_range, best_CP_curve, 'b', 'LineWidth', 1);
% mark peak values
[CP_std_max, idx_std]=max(CP_std);
plot(TSR_std(idx_std), CP_std_max, 'ko', 'MarkerFaceColor','k');
plot(best_lambda_opt, bestCp, 'bo', 'MarkerFaceColor','b');
grid on; xlabel('\lambda'); ylabel('C_P');
title('Optimized Blade vs Standard Blade: C_P(\lambda)');
legend('Standard Blade (0° pitch, 6 m/s)', sprintf('Optimized Blade
(\lambda_{design}=%.2f)', best_lambda_design), 'Std Peak', 'Optimized
Peak', 'Location', 'best');

```

Ansys WT comparison rag323.m

```

clear; clc; close all;
%Wind tunnel data: 5.5 m/s, 0 deg
lambda_WT_55=[10.02 9.95 9.88 9.58 9.48 9.29 8.99 8.71 8.44 8.19
7.87 7.57 7.25 6.83 6.63 6.48 6.30 6.11 5.81 5.55 5.30 4.98];
Cp_WT_55=[0.115 0.131 0.156 0.167 0.195 0.222 0.246 0.270 0.295
0.343 0.358 0.384 0.387 0.390 0.400 0.403 0.403 0.409 0.409
0.409 0.401 0.398];
%ANSYS data: 5.5 m/s, 0 deg
lambda_ANS_55=[4 5 6 7 8 9 10];
Cp_ANS = [0.1782350549 0.2245180822 0.2556051293 0.2264496509
0.1303336071 -0.1088262985 -0.2350668566];
% Plot comparison

```

```

figure; hold on; grid on;
plot(lambda_WT_55, Cp_WT_55, 'o-', 'LineWidth', 1, 'MarkerSize', 3);
plot(lambda_ANS_55, Cp_ANS, 'o-', 'LineWidth', 1, 'MarkerSize', 3);
xlabel('\lambda (Tip-speed ratio)');
ylabel('C_P');
title('Wind Tunnel Experimental vs ANSYS Fluent Data: C_P(\lambda), U =
5.5 m/s, \theta_p = 0^\circ');
legend('Wind Tunnel (5.5 m/s, 0^\circ)', 'ANSYS CFD (5.5 m/s,
0^\circ)', 'Location', 'best');
hold off;
%%6m/s data comparison
% Wind Tunnel Data: 0 deg, 6 m/s
lambda_wt_6=[9.06,8.81,8.63,8.29,8.08,7.83,7.51,7.17,6.75,6.48,6.14,5.73,
5.28,4.91];
Cp_wt_6=[0.266,0.289,0.309,0.336,0.353,0.373,0.390,0.405,0.418,0.424,0.43
4,0.425,0.417,0.403];
% ANSYS Data: 0 deg, 6 m/s
lambda_ansys_6= [5,6,7,8,9];
Cp_ansys_6= [0.2086,0.2416,0.2521,0.1785,0.0260];
figure; hold on; grid on;
plot(lambda_wt_6, Cp_wt_6, 'o-', 'LineWidth', 1, 'MarkerSize', 3);
plot(lambda_ansys_6, Cp_ansys_6, 'o-', 'LineWidth', 1, 'MarkerSize', 3);
xlabel('\lambda'); ylabel('C_P');
title('Wind Tunnel Experimental vs ANSYS Fluent Data: C_P(\lambda), U = 6
m/s, \theta_p = 0^\circ');
legend('Wind Tunnel Data', 'ANSYS Simulation', 'Location', 'best');
TSR_4deg=[5 6 7 8];
Cp_4deg=[0.3088 0.2933 0.1819 0.009];
% Add to the existing figure
figure(2); hold on;
plot(TSR_4deg, Cp_4deg, 'o-', 'LineWidth', 1, 'MarkerSize', 3);
legend('Wind Tunnel 0^\circ (6 m/s)', 'ANSYS 0^\circ (6 m/s)', 'ANSYS 4^\circ (6
m/s)', 'Location', 'best');
title('Comparison of Wind Tunnel vs. ANSYS CFD (0^\circ and 4^\circ Pitch, 6
m/s)');
xlabel('TSR'); ylabel('C_P'); grid on;

```

Appendix B: CFD Results Raw Data Plots

Sheet Link:

<https://docs.google.com/spreadsheets/d/1WwBMBasHeeZNYM-NHC3iZ3QRpX3J3uQLgDOnp-qxcywU/edit?usp=sharing>